Simulation Revisit

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Contents

Data Generation

To generate $(X^T, W^T, A)^T$, we first simulated 16 random variables from a zero-mean multivariate normal distribution with a first-order autoregressive covariance structure. The covariance between the k-th and l-th random variables was set to $3(0.4)^{|k-l|}$. The first 10 random variables were used as the features, X, for model training, and the next 5 as auxiliary covariates, W, for validation. The final random variable was used to obtain the protected attribute, A, and set to 1 if the normally generated value exceeded 0.253 and set to 0 otherwise. The outcome Y was generated under two different scenarios to induce varying degrees of misspecification into the ML and imputation models:

- Scenario 1: Logistic activation function with non-linear inputs

$$P(Y = 1|X, W, A = a) = expit\{-2.3 + \beta_a^T X + \gamma_a^T W + 0.2X_2^2 - 0.1X_3^2 + 0.1X_5X_6\}$$

- Scenario 2: Non-logistic activation function with non-linear hidden layer

$$P(Y = 1|X, W, A = a) = exp\{-(1.3 + \beta_a^T X + \gamma_a^T W)^2\}$$

- Scenario 3: Log-log activation function with non-monotone effect

$$P(Y = 1|X, W, A = a) = 1 - exp\{-exp(g_a(Z))\}$$

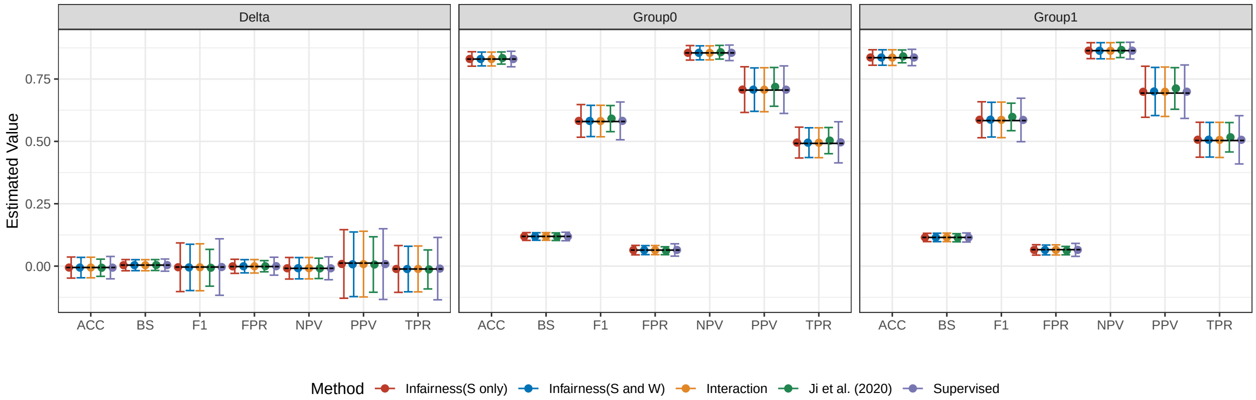
where

$$Z = -0.5 + \beta_a^T X + \gamma_a^T W + 0.1X_1W_1 - 0.1X_2X_5 + 0.12X_3^2 - 0.08X_3^3/3 + 0.10W_1^2$$

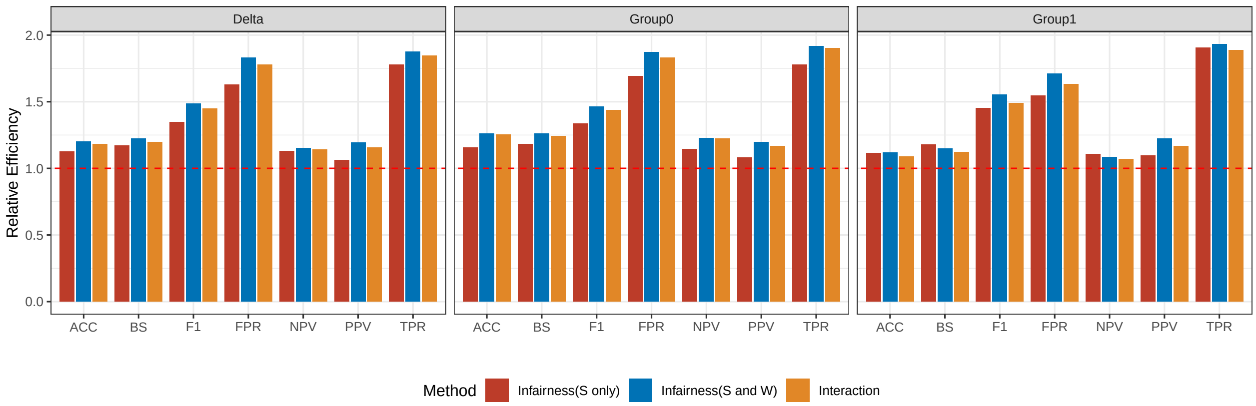
and

$$g_a(z) \;=\; b_a\, z + u_{1a}\, \exp\Big\{-\tfrac{(z+0.30)^2}{2\, s_0^2}\Big\} + d_a\, \exp\Big\{-\tfrac{z^2}{2\, s_0^2}\Big\} + u_{2a}\, \exp\Big\{-\tfrac{(z-0.30)^2}{2\, s_0^2}\Big\} + r_a\, \exp\big\{-(z/0.50)^2\big\}\, \sin\big(2\pi k\, \sigma(z)\big)$$

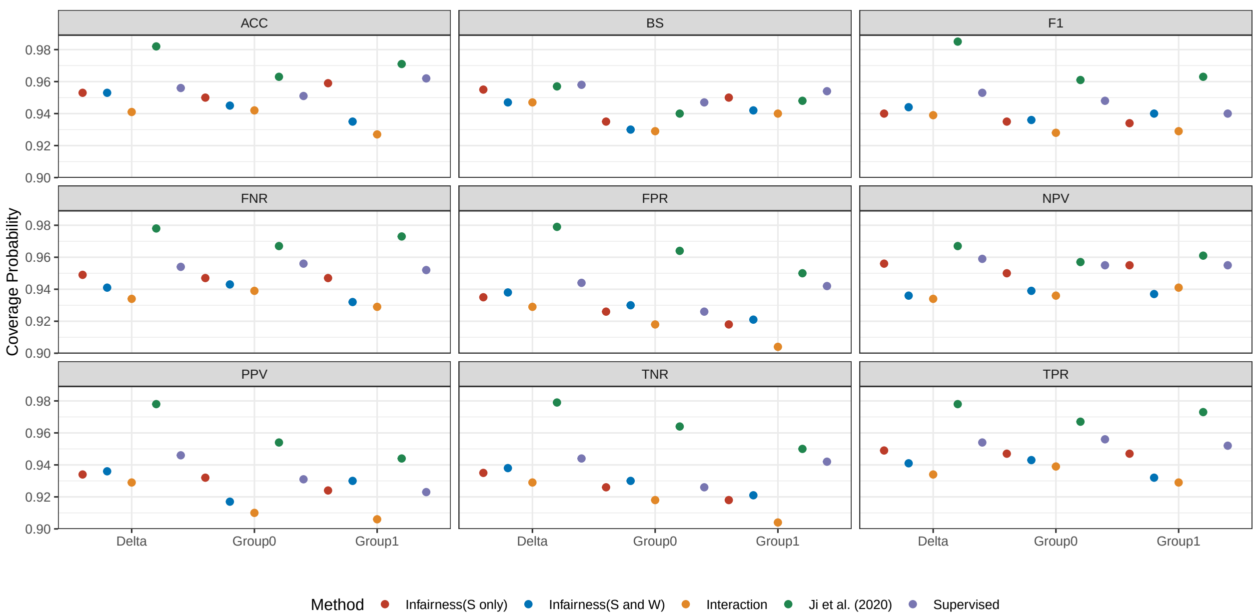
Scenario 1



Delta
Group0
Group1
Estimated Value
0.75
0.50
0.25
0.00
ACC BS F1 FPR NPV PPV TPR
Method
Infairness(S only)
Infairness(S and W)
Interaction
Ji et al. (2020)
Supervised

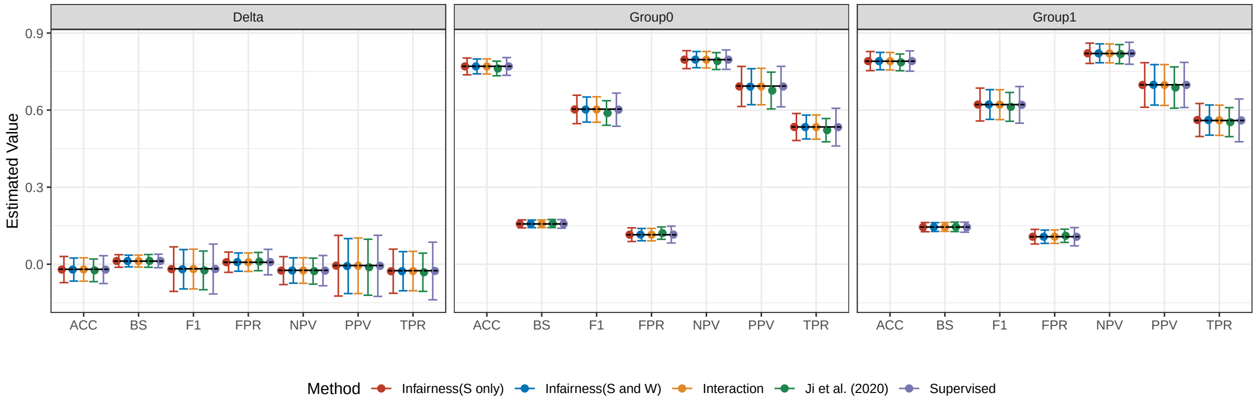


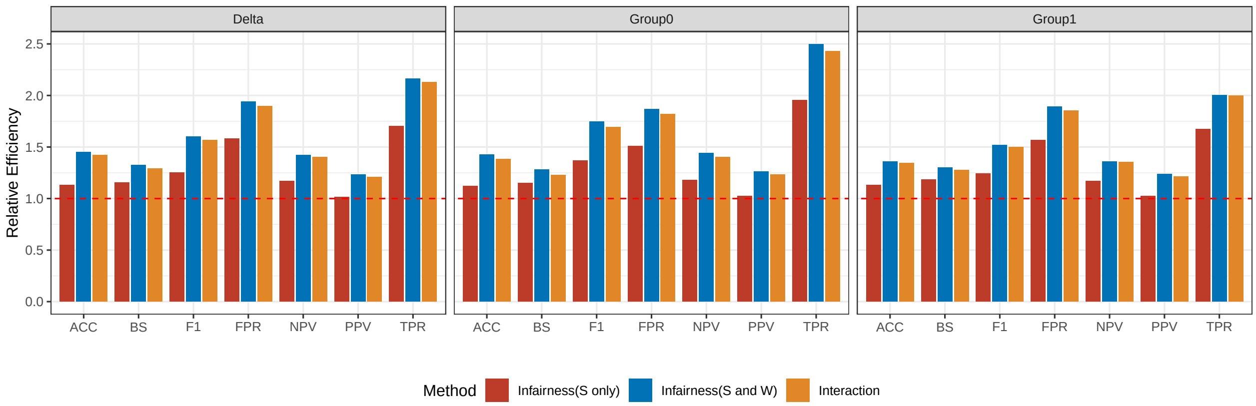
Delta
Group0
Group1
Relative Efficiency
2.0
1.5
1.0
0.5
0.0
ACC BS F1 FPR NPV PPV TPR
Method
Infairness(S only)
Infairness(S and W)
Interaction

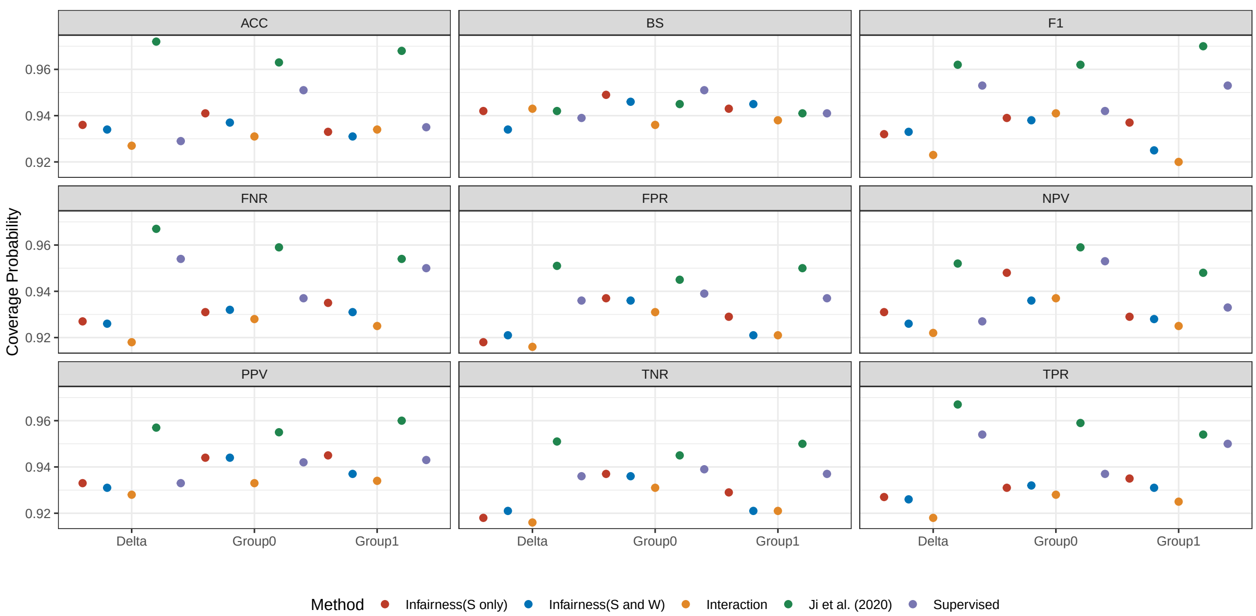


ACC
BS
F1
FNR
FPR
NPV
PPV
TNR
TPR
Coverage Probability
0.98
0.96
0.94
0.92
0.90
Delta
Group0
Group1
Method
Infairness(S only)
Infairness(S and W)
Interaction
Ji et al. (2020)
Supervised

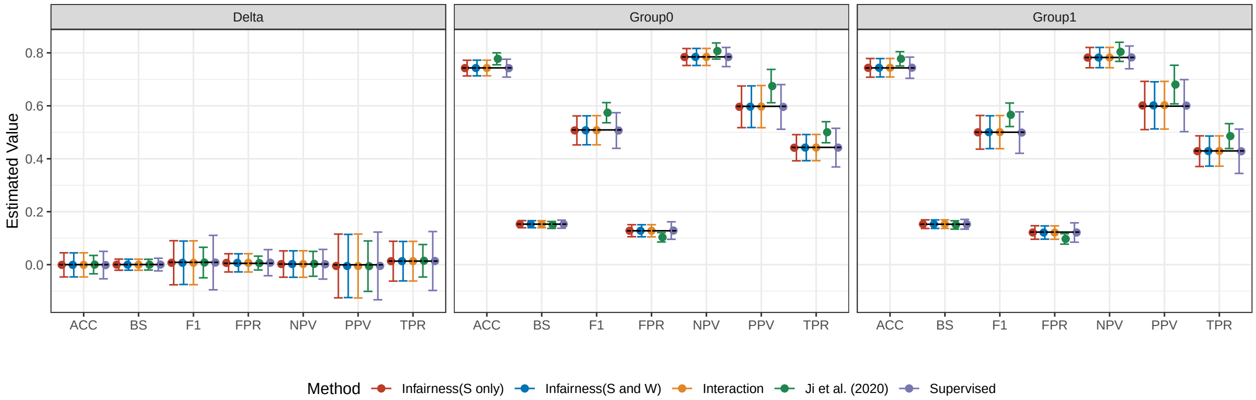
Scenario 2


Delta
Group0
Group1
Estimated Value
0.9
0.6
0.3
0.0
ACC BS F1 FPR NPV PPV TPR
Method Infairness(S only) Infairness(S and W) Interaction Ji et al. (2020) Supervised

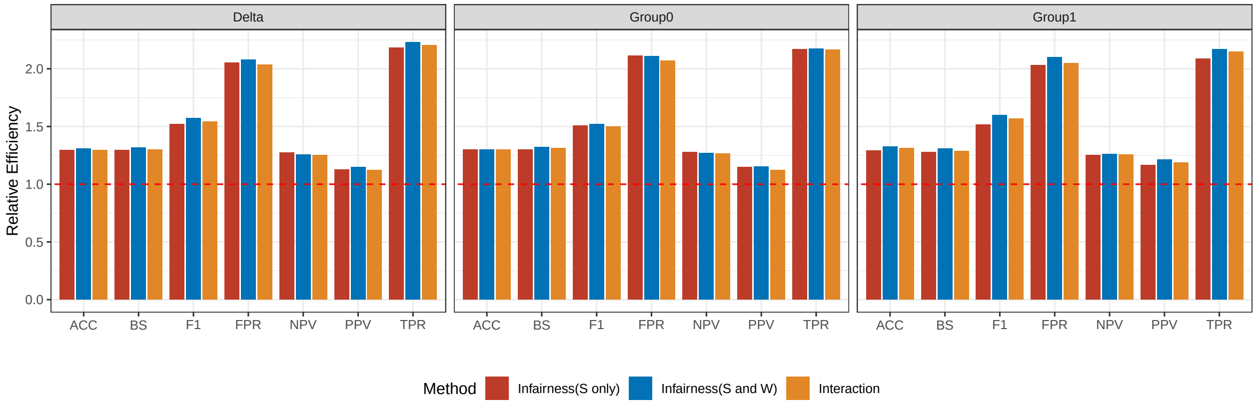

Delta
Group0
Group1
Relative Efficiency
2.5
2.0
1.5
1.0
0.5
0.0
ACC BS F1 FPR NPV PPV TPR
Method Infairness(S only) Infairness(S and W) Interaction


ACC
BS
F1
FNR
FPR
NPV
PPV
TNR
TPR
Coverage Probability
0.96
0.94
0.92
Delta Group0 Group1
Method Infairness(S only) Infairness(S and W) Interaction Ji et al. (2020) Supervised

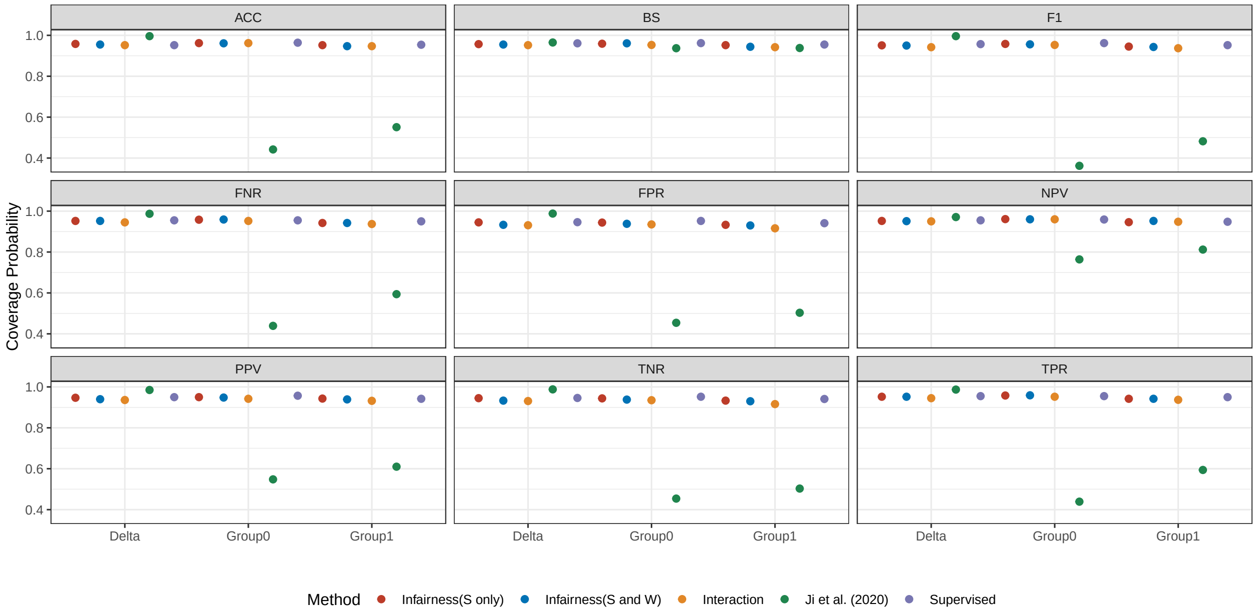
Scenario 3



Delta
Group0
Group1
Estimated Value
ACC
BS
F1
FPR
NPV
PPV
TPR
Method
Infairness(S only)
Infairness(S and W)
Interaction
Ji et al. (2020)
Supervised

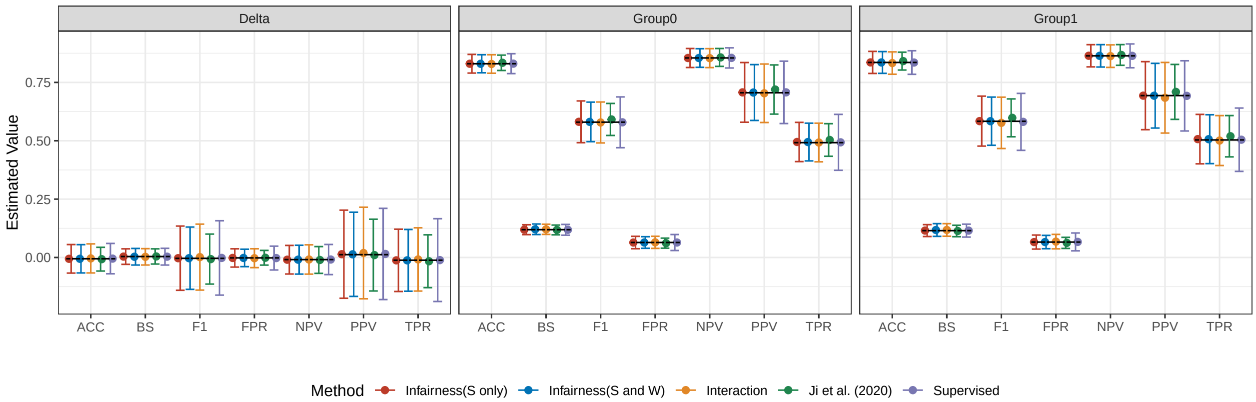


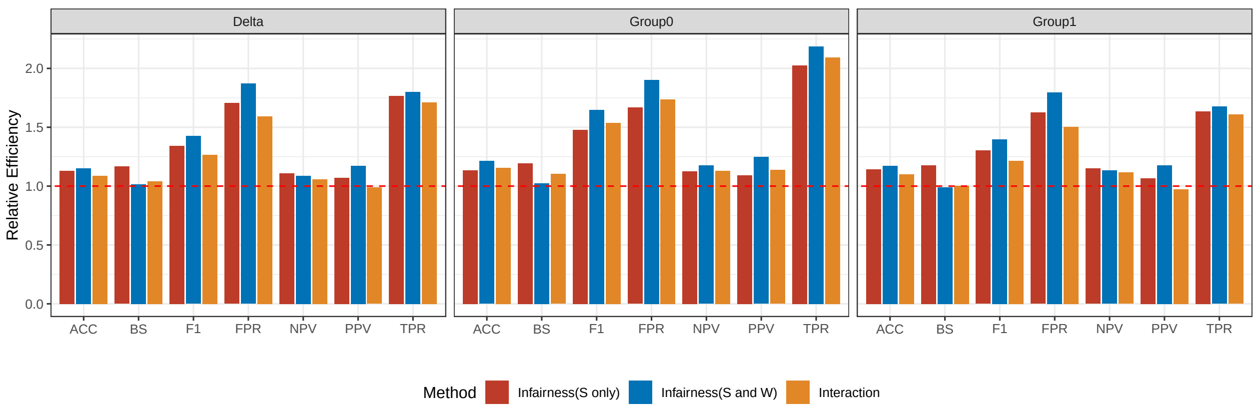
Delta
Group0
Group1
Relative Efficiency
ACC
BS
F1
FPR
NPV
PPV
TPR
Method
Infairness(S only)
Infairness(S and W)
Interaction

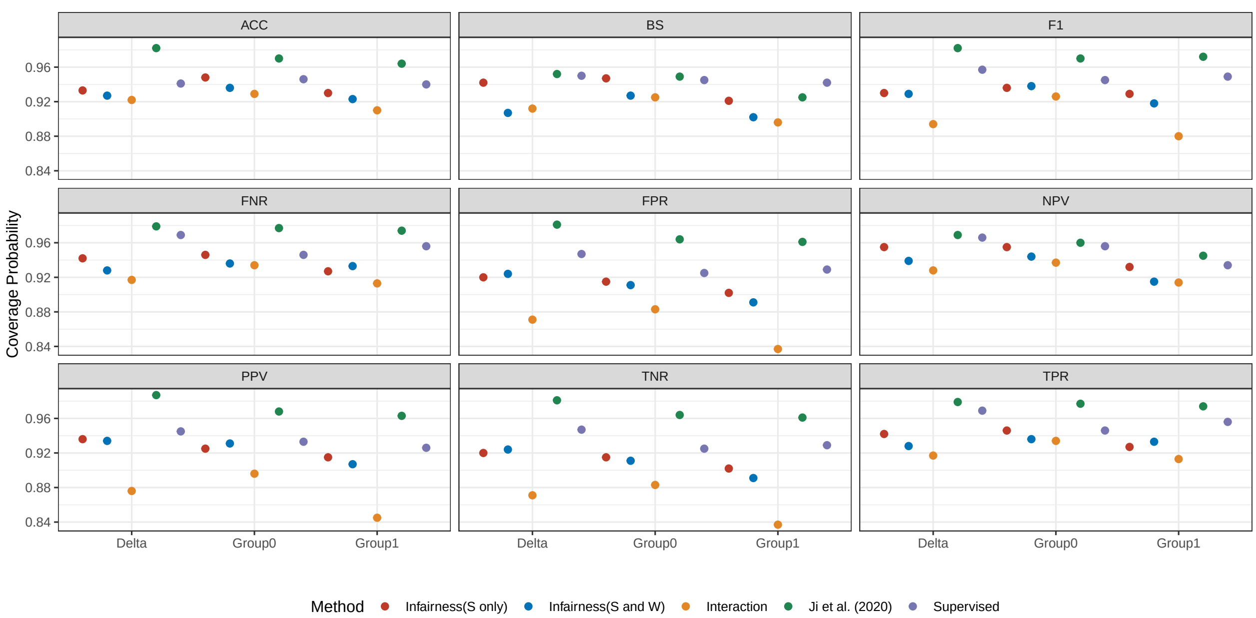


ACC
BS
F1
FNR
FPR
NPV
PPV
TNR
TPR
Coverage Probability
Delta
Group0
Group1
Method
Infairness(S only)
Infairness(S and W)
Interaction
Ji et al. (2020)
Supervised

Scenario 1 $(n = 500)$







Scenario 2 (n = 500)